

URL Shortener Service

1. Project Title

URL Shortener Service

2. Project Overview

A URL Shortener Service converts long web addresses into shorter, easy-to-share links.

Example:

Original URL:

Plain text

<https://www.example.com/products/electronics/mobile-phones/samsung-galaxy-s25-ultra>

Short URL:

Plain text

<https://short.ly/aB12X>

When users click the short URL, they are redirected to the original website.

Popular examples include Bitly and TinyURL.

3. Problem Statement

Long URLs can be:

Difficult to remember

Hard to share

Unattractive in messages

Difficult to track

A URL Shortener solves these problems by creating compact links and providing click analytics.

4. Objectives

Convert long URLs into short URLs.

Redirect users to the original URL.

Track link usage.

Provide URL analytics.

Manage custom short links.

5. Scope of the Project

The system allows users to:

URL Management

Create Short URLs

View Original URLs

Delete URLs

Analytics

Track Click Count

Track Creation Date

View Popular Links

User Management (Optional)

User Registration

User Login

Personal URL Dashboard

6. System Architecture

Plain text

Client

|



REST API Server

|



Business Logic

|



Database

Components

Client

Sends URL shortening requests

Receives short URLs

API Server

Generates unique short codes

Handles redirection

Database

Stores URL information

7. Functional Requirements

URL Creation Module

Users submit a long URL.

Input:

Original URL

Output:

Short URL

Example:

Input:

Plain text

<https://www.example.com/about/company/history>

Output:

Plain text

<https://short.ly/abc123>

URL Redirection Module

When a user clicks a short URL:

Plain text

<https://short.ly/abc123>

The system redirects to:

Plain text

<https://www.example.com/about/company/history>

Analytics Module

Track:

Number of Clicks

URL Creation Date

Last Access Date

Custom URL Module

Users can create custom links.

Example:

Plain text

<https://short.ly/myproject>

instead of:

Plain text

<https://short.ly/xY8kL>

8. Non-Functional Requirements

Security

URL validation

Protection against malicious links

Performance

Fast redirection

Reliability

Accurate URL mapping

Scalability

Support millions of URLs

9. Database Design

Users Table (Optional)

Field

Type

user_id

Integer

name

Varchar

email

Varchar

password

Varchar

URLs Table

Field

Type

id

Integer

original_url

Text

short_code

Varchar

click_count

Integer

created_at

DateTime

expires_at

DateTime

10. API Endpoints

Create Short URL

Http

POST /api/shorten

Request:

JSON

```
{  
  "url": "https://www.example.com"  
}
```

Response:

JSON

```
{  
  "shortUrl": "https://short.ly/abc123"
```

}

Redirect URL

Http

GET /:shortCode

Example:

Http

GET /abc123

System redirects to original URL.

Get URL Analytics

Http

GET /api/analytics/:shortCode

Delete URL

Http

DELETE /api/url/:id

11. Technology Stack

Backend

Node.js

Framework

Express.js

Database

MongoDB

Unique ID Generator

Nano ID

Testing Tool

Postman

12. Project Workflow

Step 1

User submits a long URL.

Step 2

System validates the URL.

Step 3

System generates a unique short code.

Example:

Plain text

abc123

Step 4

System stores:

Original URL

Short Code

in the database.

Step 5

Short URL is returned.

Example:

Plain text

<https://short.ly/abc123>

Step 6

User opens the short URL.

Step 7

System finds the original URL.

Step 8

User is redirected to the destination website.

Step 9

Click count is updated.

13. Sample API Request

Create Short URL

JSON

```
{  
  "url": "https://www.google.com"  
}
```

Response

JSON

```
{  
  "message": "URL Shortened Successfully",  
  "short Url": "https://short.ly/abc123"  
}
```

14. Advantages

Easy URL sharing

Cleaner links

Click tracking

Better user experience

Useful for marketing campaigns

15. Future Enhancements

QR Code Generation

Generate QR codes for each short URL.

User Dashboard

Manage all shortened URLs.

Password-Protected Links

Restrict access to links.

URL Expiration

Automatically disable links after a set date.

Analytics Dashboard

Show:

Daily clicks

Monthly clicks

Geographic statistics

Custom Domains

Example:

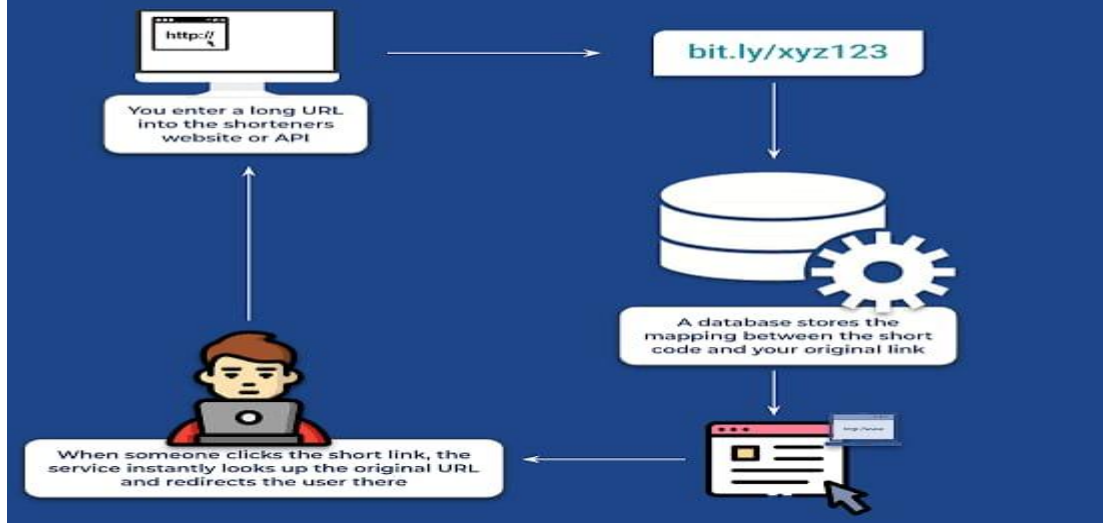
Plain text

<https://mycompany.in/sale2026>

16. Project Conclusion

The URL Shortener Service is a backend project that converts long URLs into compact, shareable links while providing redirection and analytics features. It demonstrates key backend concepts such as REST APIs, database management, routing, unique ID generation, URL validation, analytics tracking, and scalable system design, making it an excellent project for backend development portfolios and academic submissions.

Let's walk through the journey of a single click:



How URL Shorteners Work

